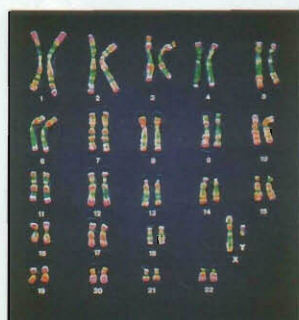


GENETICS



EACH PERSON IS UNIQUE, but he or she also inherits some characteristics and even appearance from his or her parents. The study of how characteristics are passed on from parents to offspring is known as genetics, and it affects all forms of life. At the centre of the process is the deoxyribonucleic acid (DNA) molecule, which exists inside every living cell and contains a complex chemical "code" that controls the way in which life forms are put together and operate. DNA is composed of genes, and DNA, in turn, makes up chromosomes. All of these microscopic structures live in the nuclei of cells.



Chromosomes

A chromosome is a thread-like structure found in the nucleus of a cell. Chromosomes store DNA and carry DNA molecules when a cell reproduces by dividing. Most human cells contain 46 chromosomes, divided into 23 pairs; 23 chromosomes are derived from each of the parents.

Chromosomes, stained and paired

Chromosome defects

A chromosome defect can result from the wrong number of chromosomes, a missing piece of chromosome, or an unnecessary extra piece. Alternatively, there may be a "mistake" in part of the DNA. Any of these can cause a genetic disorder before or after birth, or later in life.

Genetic defect causes malformed feet.



Genes

A gene is the basic unit of inheritance, a small segment of a DNA molecule. There are about 100,000 genes in the 46 human chromosomes. Genes contain the instructions to construct proteins, which control cell activities. Therefore, genes help to determine the characteristics of an organism.



Grey-blue eye, round-shaped, with long eyelashes, and large eyelid



Dark brown eye, almond-shaped, with short eyelashes



Medium-brown eye with fine eyelashes, and small eyelid

Genotype and phenotype

The genotype is the overall genetic blueprint for an organism. The phenotype is what an organism actually looks like, based on genotype and environment.

Alleles

Each gene has two or more forms called alleles. They control the same characteristic (for example, eye colour) but different versions of it (for instance, blue or brown).

DNA

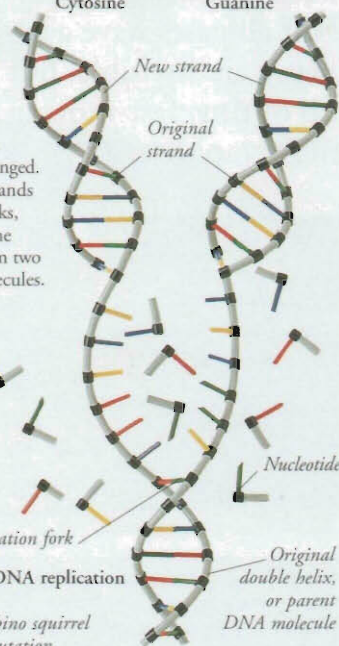
A DNA molecule contains all the information required to make and operate a specific organism. DNA is found in the nucleus of a cell and is a long structure which consists of two strands twisted together to form a double helix. The strands are linked by four chemicals called bases: thymine, adenine, cytosine, and guanine.

"Backbone" of strand is made from sugar and phosphate molecules.



Replicating DNA

DNA is unique in its ability to replicate, or copy, itself. When a cell divides (reproduces), DNA information is passed on unchanged. During replication, the DNA strands separate, and DNA building blocks, nucleotides, line up, matching the original sequence of bases to form two new and identical DNA molecules.



Mutations

A mutation is an accidental change in the structure of part of a DNA molecule, or sometimes in the number or shape of chromosomes. Mutations may result in new or unusual characteristics.



Variations

While members of a species look similar, they are not identical, as we can see from people around us.

Variation happens because each individual receives a unique combination of DNA from its parents during reproduction.

Variations in types of flowers



Franklin, Watson, and Crick

The discovery of the DNA molecule was a collaborative effort. Rosalind Franklin (1920–58) completed groundbreaking work, which was consolidated in 1953 by Francis Crick (b.1916), James Watson (b.1928), and Maurice Wilkins (b.1916). Watson, Crick, and Wilkins shared the 1962 Nobel Prize.



Rosalind Franklin, British biochemist



James Watson, American biologist



Francis Crick, British scientist

A length of DNA molecule